

Cambridge International AS & A Level

FURTHER MATHEMATICS**9231/32**

Paper 3 Further Mechanics

October/November 2024

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **19** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

PUBLISHED**Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
 - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
 - The total number of marks available for each question is shown at the bottom of the Marks column.
 - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
 - Square brackets [] around text or numbers show extra information not needed for the mark to be awarded.

Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1	Hooke's law: $T = \frac{100}{0.8}x$	B1	$T = \frac{100}{0.8}(r - 0.8)$
	N2L: $T = 2 \times (0.8 + x) \times 5^2$	B1	$T = 2 \times r \times 5^2$
	Equate and solve: $50(0.8 + x) = 125x$, $x = 0.533$	B1	$\frac{8}{15}$
		3	

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Question	Answer	Marks	Guidance
2	Extension when P comes to rest is x , EPE loss = $\frac{1}{2} \times \frac{5mg}{a} \times \left(\left(\frac{a}{2} \right)^2 - x^2 \right)$	B1	Both terms seen.
	Energy: $\frac{1}{2} \times \frac{5mg}{a} \times \left(\left(\frac{a}{2} \right)^2 - x^2 \right) = mg \left(\frac{a}{2} - x \right)$	M1	At least one EPE term and a GPE term, dimensionally correct.
	Solve: $20x^2 - 8ax - a^2 = 0$	M1	Obtain homogeneous quadratic equation in x and a Must come from an energy equation involving two EPE terms. Note that the correct case simplifies to a linear equation $\frac{5mg}{2a} \left(\frac{a}{2} + x \right) = mg$ and this scores M1.
	$x = -\frac{a}{10}$, so distance of P below O is $\frac{9}{10}a$	A1	

Question	Answer	Marks	Guidance
2	Alternative method for question 2		
	Distance of P below O when it comes to rest is h EPE loss = $\frac{1}{2} \times \frac{5mg}{a} \times \left(\left(\frac{a}{2} \right)^2 - (h-a)^2 \right)$	B1	
	Energy: $\frac{1}{2} \times \frac{5mg}{a} \times \left(\left(\frac{a}{2} \right)^2 - (h-a)^2 \right) = mg \left(\frac{3}{2}a - h \right)$	M1	At least one EPE term and a GPE term, dimensionally correct.
	$20h^2 - 48ah + 27a^2 = 0$	M1	Obtain homogeneous quadratic/linear equation in h and a . Must come from an energy equation involving two EPE terms.
	$(10h - 9a)(2h - 3a) = 0, \quad h = \frac{9}{10}a$	A1	
		4	

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Question	Answer	Marks	Guidance
3(a)	PCLM along line of centres: $-mv = m2u \cos \theta - m3u \sin \theta$	B1	Must include m , must have minus sign on RHS, accept positive or negative v . If velocity of B after collision is included, it must be equated to zero before this mark is awarded.
	NEL: $v = eu(3 \sin \theta + 2 \cos \theta)$	M1	Must have plus sign on RHS, accept positive or negative v (sign of v does not need to be consistent with PCLM equation) If velocity of B after collision is included, it must be equated to zero before this mark is awarded.
	Eliminate v : $6 \sin \theta = 8 \cos \theta$, $\tan \theta = \frac{4}{3}$	A1	Correct work only, except possibly missing m .
		3	

Question	Answer	Marks	Guidance
3(b)	Only change in KE is along line of centres $\text{Loss} = \frac{1}{2}m\left((2u\cos\theta)^2 + (3u\sin\theta)^2\right) - \frac{1}{2}mv^2$	M1	
	$\frac{1}{2}mu^2\left(\frac{36}{25} + \frac{144}{25} - \left(\frac{12}{5} - \frac{6}{5}\right)^2\right) = \frac{72}{25}mu^2$	A1	(Note that $\tan\theta = \frac{2}{3}$ leads to final answer $\frac{36}{13}mu^2$)
		2	
	Alternative method for question 3(b)		
	Alternative method, using total KE $\text{Loss in KE} = \left[\frac{1}{2}m(2u)^2 + \frac{1}{2}m(3u)^2 \right] - \left[\frac{1}{2}mv^2 + \frac{1}{2}m(2u\sin\theta)^2 + \frac{1}{2}m(3u\cos\theta)^2 \right]$	M1	Or equivalent, with all necessary terms present
	$\frac{13}{2}mu^2 - \frac{181}{50}mu^2 = \frac{72}{25}mu^2$	A1	
3(c)		2	
	[Components of velocity of A after collision are $\leftarrow \frac{6}{5}u$ $\downarrow \frac{8}{5}u$ so] angle between line of centres and A's direction is θ .	M1	
	Angle of deflection = $180^\circ - 2\tan^{-1} 4/3 = 73.7^\circ$	A1FT	FT their answer to part (a)
		2	

Question	Answer	Marks	Guidance
4(a)	In this question, allow equivalent marks for resolutions in different directions and moments about other points. Apply the guidance given in the main scheme. $\uparrow T \cos \theta + F = 2W$	B1	
	$\rightarrow T \sin \theta = R$	B1	
	Moments about A: $T \cos \theta \times 3a \sin \theta + T \sin \theta \times 3a \cos \theta = W \times 3a \sin \theta + W \times ka \sin \theta$ OR Moments about C: $R \times 3a \cos \theta = F \times 3a \sin \theta + W \times (ka - 3a) \sin \theta$	M1	All relevant terms included, dimensionally correct, forces must be resolved if appropriate. Allow sin/cos mix, allow sign errors. LHS: any equivalent expression, for example $3Ta \sin 2\theta$, $3T \sin(180 - 2\theta)$. All relevant terms included, dimensionally correct, forces must be resolved if appropriate. Allow sin/cos mix, allow sign errors.
	$[6aT \cos \theta = (3+k)aW \text{ and } \frac{T}{3}(3 \cos \theta + \sin \theta) = 2W \text{ give}] \text{ [and give and give]}$ $12 \cos \theta = (3+k)(\cos \theta + \frac{1}{3} \sin \theta)$	M1	Use $F = \frac{1}{3}R$ and eliminate T and W to obtain an expression in k and θ , dependent on a dimensionally correct moments equation.
	$k = 5$	A1	
		5	

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Question	Answer	Marks	Guidance
4(b)	A complete method to find F in terms of W	M1	Any complete method to find F . For example, substitute into moments equation to obtain T [$T = \frac{2}{3}\sqrt{13} W$, $R = 2W$].
	$F = \frac{2}{3}W$	A1	Correct.
		2	

Question	Answer	Marks	Guidance
5(a)	Use correct equation of trajectory: $\frac{4}{5}a = 3a \times \frac{1}{3} - \frac{g}{2u^2} \times (3a)^2 \times \left(1 + \frac{1}{9}\right)$	M1	No (implied) sight of trajectory equation M0.
	$\frac{4}{5}a = a - \frac{5ga^2}{u^2}, \frac{5ga}{u^2} = \frac{1}{5}, u^2 = 25ga$	A1	At least one step of intermediate working must be seen. AG
		2	

Question	Answer	Marks	Guidance
5(b)	For P , time of flight T and range R For Q , time of flight $\frac{1}{2}T$ and range R $\left[T = \frac{2u \sin \theta}{g} = \sqrt{\frac{10a}{g}} \right]$	B1	Time of flight for P or Q .
	[From motion of P , $R = \frac{2}{g} \times 25ag \times \frac{3}{10} = 15a$]	B1	Range for P .
	For Q : $\rightarrow R = v \cos \alpha \times \frac{1}{2}T$, $v \cos \alpha = \frac{2R}{T}$	M1	Obtain an expression for $v \cos \alpha$. May involve u and θ .
	$\uparrow 0 = v \sin \alpha \times \frac{1}{2}T - \frac{1}{2}g \left(\frac{1}{2}T \right)^2$, $v \sin \alpha = \frac{1}{4}gT$	M1	Obtain an expression for $v \sin \alpha$. May involve u and θ .
	Square and add: $v^2 = \left(\frac{2R}{T} \right)^2 + \left(\frac{1}{4}gT \right)^2 \left[= 90ag + \frac{5}{8}ag \right]$	M1	
	$v^2 = \frac{725}{8}ag$	A1	$\tan \alpha = \frac{\frac{1}{4}gT}{\frac{2R}{T}} = \frac{1}{12}$

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Question	Answer	Marks	Guidance
5(b)	Alternative method for question 5(b)		
	For P , time of flight T and range R For Q , time of flight $\frac{1}{2}T$ and range R Horizontal motion for P and Q $R = u \cos \theta T$ and $R = (v \cos \alpha) \frac{T}{2}$	M1	Both.
	Vertical motion for P and Q $u \sin \theta = \frac{gT}{2}$ and $v \sin \alpha = \frac{gT}{4}$	M1	Both, may come from using $s = ut + \frac{1}{2}at^2$.
	Equate two expressions for R : $v \cos \alpha = 2u \cos \theta$	A1	$v \cos \alpha = \frac{6}{\sqrt{10}}u$
	Equate two expressions for vertical motion: $v \sin \alpha = \frac{1}{2}u \sin \theta$	A1	$v \sin \alpha = \frac{1}{2\sqrt{10}}u$
	Square and add: $v^2 = u^2 \left(4 \cos^2 \theta + \frac{1}{4} \sin^2 \theta \right) \left[= \frac{29}{8}u^2 \right]$	M1	
	$\frac{29}{8} \times 25ag = \frac{725}{8}ag$	A1	
		6	

Question	Answer	Marks	Guidance
6(a)	For P to lowest point L : Energy: $\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mga(1 - \cos \theta)$ $[v^2 = \frac{28}{9}ag - 2ag \cos \theta]$	M1*	Dimensionally correct, all terms present, allow sign errors, allow cos/sin error.
	From L to string goes slack: Energy: $\frac{1}{2}mv^2 = \frac{1}{2}mw^2 + \frac{4a}{9}mg(1 + \cos \theta)$ $\left[w^2 = \frac{20}{9}ag - \frac{26}{9}ag \cos \theta\right]$	M1*	Dimensionally correct, all terms present, with $\frac{4a}{9}$, allow sign errors, allow cos/sin error.
	Both equations correct, allow unsimplified.	A1	
	When string goes slack: $mg \cos \theta = \frac{mw^2}{4a}$	B1	
	Equate expressions for w^2 to find a value for $\cos \theta$	DM1	
	$\cos \theta = \frac{2}{3}$	A1	
		6	

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Question	Answer	Marks	Guidance
	Alternative method for question 6(a)		
	For P from start to string goes slack: Energy: $\frac{1}{2}mw^2 = \frac{1}{2}mu^2 + mga \left[(1 - \cos \theta) - \frac{4}{9}(1 + \cos \theta) \right]$	M2	Dimensionally correct, all terms present, with $\frac{4}{9}a$ Allow sign errors, allow cos/sin error RHS may appear with $\frac{5}{9}a - \left(1 + \frac{4}{9}\right)a \cos \theta$ Allow M1 if $\frac{4}{9}$ is missing or if an attempt at both heights, but all other conditions are met.
	$\left[w^2 = \frac{10}{9}ag + 2ag \left(\frac{5}{9} - \frac{13}{9} \cos \theta \right) = \frac{20}{9}ag - \frac{26}{9}ag \cos \theta \right]$	A1	Correct, allow unsimplified.
	When string goes slack: $mg \cos \theta = \frac{mw^2}{4a}$	B1	
	Equate expressions for w^2 to find a value for $\cos \theta$	DM1	
	$\cos \theta = \frac{2}{3}$	A1	
		6	

Question	Answer	Marks	Guidance
6(b)	Tension before: $T_1 - mg = \frac{mv^2}{a}$ Tension after: $T_2 - mg = \frac{mv^2}{\frac{4}{9}a}$	M1	EITHER equation, dimensionally correct, allow sign error only
	$\left[v^2 = \frac{16}{9}ag, \right] \quad T_1 = \frac{25}{9}mg \quad T_2 = 5mg$	A1	EITHER tension correct
	Find the other tension (from a valid equation) and find ratio of tensions	M1	Equation must be of the form $T - mg = \frac{mv^2}{r}$
	Ratio is 5 : 9	A1	Any equivalent ratio, allow $\frac{5}{9}$
		4	

Question	Answer	Marks	Guidance
7(a)	$m \frac{dv}{dt} = mg - 0.1mv^2$	B1	Use of suvat means 0 marks in this part Note that no mg term means 0 marks in this part. Must see m , may be cancelled before $a = \frac{dv}{dt}$ used
	$\frac{dv}{dt} = 10 - 0.1v^2 = \frac{1}{10}(100 - v^2)$ $\frac{dv}{100 - v^2} = \frac{1}{10} dt$ $\ln \left \frac{v+10}{10-v} \right = 2t + A$	M1*	Separate variables and integrate. May see partial fractions, but integral is on Formula sheet, allow missing $+A$ for M1 only
		A1	Must see modulus sign
	Use $t = 0, v = 0, A = 0$	DM1	
	Remove logs to obtain v in terms of t	M1	
	$v = \frac{10(e^{2t} - 1)}{e^{2t} + 1}$ aef	A1	$v = \frac{10(1 - e^{-2t})}{1 + e^{-2t}}$
		6	

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Question	Answer	Marks	Guidance
7(b)	$v \frac{dv}{dx} = \frac{1}{10}(100 - v^2)$	M1*	Use of suvat means 0 marks in this part Separate variables and integrate, allow missing +A for M1 only.
	$-\frac{1}{2} \ln(100 - v^2) = \frac{1}{10}x + B$	A1	For A1, allow missing modulus sign
	Use $x=0, v=0, B = -\frac{1}{2} \ln 100$	DM1	
	Remove logs to obtain v^2 in terms of x	M1	
	$v^2 = 100(1 - e^{-\frac{x}{5}})$	A1	AEF Allow 10g instead of 100.
		5	